Fiscal Year:	FY 2020	**Task Last Updated:**	FY 09/03/2019
PI Name:	Sekulic, Dusan D.Sc.		
Project Title:	BRAzing of Aluminum Alloys IN Space		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Materials science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	40506	**Congressional District:**	6
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2015 NNH15ZTT002N MaterialsLab Open Science Campaigns for Experiments on the International Space Station
Start Date:	11/04/2016	**End Date:**	11/03/2020
No. of Post Docs:	0	**No. of PhD Degrees:**	1
No. of PhD Candidates:	3	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA MSFC
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Flight Program:	ISS		
Flight Assignment:			
Key Personnel Changes/Previous PI:	September 2019: No changes		
COI Name (Institution):	Krivilev, Mikhail Ph.D. (Udmurt State University, Russia) Mesarovic, Sinisa Ph.D. (Washington State University)		
Grant/Contract No.:	NNX17AB52G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	We propose to investigate the techniques of brazing (bonding by molten alloy) in microgravity for space construction and repair of spacecrafts damaged by micrometeoroids and orbital debris. A series of space (International Space Station - ISS) and terrestrial experiments are planned to be integrated with modeling and analysis. These are proposed to answer the practical and fundamental questions: (1) How can we take advantage of the absence of gravity and dominance of capillary forces in transporting the large quantity of molten braze? (2) What composition of the brazing alloy and the brazing technique will enable strong and efficient bonding of structural elements and repair? (3) Can this brazing be done using flux-less cladding? The phenomena to be studied under these conditions include molten metal kinetics, wetting and spreading of liquid clad/filler over topographically modified surfaces. The applied aspects include direct verification if brazing of Al alloys is transferable to orbit for rapid repair of the pressure wall. For the first time, brazing in space will be processed using the patented flux-less Trillium technology. Further development includes design and testing of a space-based brazing hot zone with the support from Marshall Space Flight Center (MSFC).
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The impact of the proposed research on the life on Earth manifests itself through multifaceted benefits. These include: (i) uncovering so-far hidden physical/chemical/metallurgical aspects of metal bonding by brazing (extensively used in manufacturing) through uncoupling of the gravity impact, hence transforming the art of this technology to a predictive science for terrestrial applications, (ii) exploring a possibility to use novel materials as bonding alloys less sensitive on terrestrial hindrances (e.g., oxidation) in designing bonded assemblies, (iii) featuring sustainable resources use and less energy invested due to an expected broadening the range of applicability of bonding under adverse conditions, (iv) developing visualization of the brazing process in situ and in real time under vacuum and/or controlled atmosphere, (v) developing the kinetics of wetting theory by implementing phase field techniques that can be used to model capillary phenomena not only for liquid metals but for other material systems in both micro gravity and gravity (on Earth) conditions.
Task Progress:	This document constitutes the Year 3 report of the research work conducted within the scope of the NASA sponsored “Brazing Aluminum in Space” (BRAINS) project (NNX17AB52G). The work has included both experimental and theoretical modeling segments. The participation of both collaborating teams (the University of Kentucky, UK, and the Washington State University, WSU), established in Year 1, continues. A full participation of the Russian team continues as well. Both experimental and theoretical (modeling) work have been completed as planned at UK, WSU, and UdSU (Udmurt State University). The experimental work under terrestrial conditions was focused on further populating the map of parameters indicating domains of filling or not filling the micro/macro holes by surface tension driven liquid aluminum alloys. The modeling activities during the third year continued being focused on the following tasks/issues: (1) Modeling of surface tension-controlled equilibrium formations of molten aluminum, (2) Modeling of the kinetics of the process of filling the holes by surface micro layers of liquid aluminum alloy. The experimental work in Year 3 explored both phenomenological and theoretical modeling for the hole-in-a-brazing-sheet filling problem in a controlled atmosphere brazing process. Solid alpha-Al formation during the brazing process promotes the possibility that liquid aluminum may travel over sharp edges through micro-channels. This may further support the phenomenon of changing liquid volume on the wall of the hole. The model has been further developed by using the principle of minimal potential energy (modeling proceeds by (i) further developing an in-house constructed phase-field theory and numerical simulation of the kinetics and (ii) equilibrium free surface formation by Surface Evolver.) The model fits experimental results with some deviations, due to assumptions made during the model development. In Year 3 we have considered the design of the experimental samples to be used in SUBSA (Solidification Using Baffle in Sealed Ampoules) facility. Theoretical analysis has included modeling equilibrium shapes of free surfaces of molten metal upon filling or non-filling the hole. Agreement with empirical data has been satisfactory. The phase-field simulations of hole-filling in brazing reproduce the flow behavior in agreement with the experimentally observed one.
Bibliography Type:	Description: (Last Updated: 12/06/2025)
Articles in Peer-reviewed Journals	Shutov IV, Kamaeva LV, Krivilyov MD, Yu C-N, Mesarovic SM, Sekulic DP. "Effect of processing parameters on microstructure in brazing of Al-Si alloys." Journal of Crystal Growth. 2020 Jan 15;530:125287. Available online 10 October 2019. https://doi.org/10.1016/j.jcrysgro.2019.125287 , Jan-2020
Articles in Peer-reviewed Journals	Koehler T, Yu C-N, Schmidt K, Graetzel M, Wu Y, Bergmann JP, Sekulic DP. "Development of a repair process for manmade structures in space by using aluminum brazing and friction stir spot welding." Welding Technology Review. 2019 Nov;91(9):39-50. https://doi.org/10.26628/wtr.v91i9.1072 [Note: previously reported in Aug 2019 as "In press as of August 2019"] , Nov-2019
Awards	Sekulic DP, Mesarovic SDj, Krivilev MD, Yu C-N, Wu Y, Lazaridis K, Shutov IV, Kamaev LV. "Invited Lecture: A large mass of Al-Si molten braze in a gravity field, 6th International Conference Advances in Brazing and Soldering Technologies, Wroclaw, Poland, Sept. 18-20, 2019." Sep-2019
Awards	Dehsara M, Fu H, Mesarovic SDj, Sekulic DP, Krivilyov M. "Keynote lecture: Phase field models for capillary flows. 7th European Conference on Computational Fluid Dynamics, Glasgow, UK, June 11-15, 2018." Jun-2018
Awards	Wu Y, Yu C-N, Sekulic DP. "Honorable Mention Award, Art Competition, Title: 'Dancing of Liquid Metal.' 34th Annual Meeting of the American Society for Gravitational and Space Research, Bethesda, MD, 2018." Nov-2018
Awards	Yu C-N, Wu Y, Sekulic DP. "2nd runner-up of Best Poster Award, Title: Modeling of a capillary driven cavity filling process in aluminum brazing. Mechanical Engineering Graduate Showcase: Poster Competition, University of Kentucky, Lexington, KY, USA, 2018." Nov-2018
Books/Book Chapters	Mesarovic SDj. "Physical foundations of mesoscale continua." in "Mesoscale models: From micro-physics to macro-interpretation. CISM International Centre for Mechanical Sciences book series." Ed. S.Dj. Mesarovic, S. Forest, H.M. Zbib. Cham : Springer, 2019. CISM International Centre for Mechanical Sciences (Courses and Lectures), vol 587, p. 1-50. https://doi.org/10.1007/978-3-319-94186-8_1 . , Jan-2019

Books/Book Chapters	Sekulic DP. "Brazing of Aluminum Alloys." in "ASM Handbook – Aluminum Science and Technology." Ed. K. Anderson, J. Weritz, J.G. Kaufman. Materials Park, OH : ASM International, 2018. Vol. 2A, p. 763-782. ISBN: 978-1-62708-207-5. https://doi.org/10.31399/asm.hb.v02a.a0006529 , Nov-2018
Papers from Meeting Proceedings	Mesarovic SDj, Sekulic DP, Krivilyov M, Shutov I, Voroshilov E. "Wetting phenomena in brazing of Al alloys in microgravity (space experiment BRAINS)." International Conference “Solidification: computer simulation, experiments and technology 2019,” Izhevsk, Russia, April 11-12, 2019. Proceedings of the Udmurt State University Condensed Matter Lab, Udmurt Federal Research Center of the Ural Branch of the Russian Academy of Sciences, 2019. , Apr-2019